Neural Computation -- 1999

107 article(s), all issues

1. Combined 5 2 cv F Test for Comparing Supervised Classification Learning Algorithms

Alpayd■nEthem

https://doi.org/10.5555/1121924.1121932

Dietterich (1998) reviews five statistical tests and proposes the 5 2 cv t test for determining whether there is a significant difference between the error rates of two classifiers. In our experime...

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2. Neuronal Regulation: A Mechanism for Synaptic Pruning During Brain Maturation

ChechikGal, MeilijsonIsaac, RuppinEytan

https://doi.org/10.5555/1121924.1121928

Human and animal studies show that mammalian brains undergo massive synaptic pruning during childhood, losing about half of the synapses by puberty. We have previously shown that maintaining the ne...

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3. Detecting and Estimating Signals in Noisy Cable Structures, I: Neuronal Noise Sources

ManwaniAmit, KochChristof

(no DOI)

In recent theoretical approaches addressing the problem of neural coding, tools from statistical estimation and information theory have been applied to quantify the ability of neurons to transmit i...

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4. Natural Gradient Learning for Over-and Under-Complete Bases in ICA

AmariShun-ichi

https://doi.org/10.5555/1121924.1121931

Independent component analysis or blind source separation is a new technique of extracting independent signals from mixtures. It is applicable even when the number of independent sources is unknown...

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5. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1999.11.issue-5

No abstract available.

6. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1999.11.issue-4

No abstract available.

7. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1999.11.issue-2

No abstract available.

8. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1999.11.issue-8

No abstract available.

9. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1999.11.issue-1

No abstract available.

10. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1999.11.issue-7

No abstract available.

11. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1999.11.issue-6

No abstract available.

12. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.1999.11.issue-3

No abstract available.

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13. Evolution of Time Coding Systems

Catherine Carr, M. Friedman

https://doi.org/10.1162/089976699300016773 | pp. 1-20

The auditory and electrosensory systems contain circuits that are specialized for the encoding and processing of microsecond time differences. Analysis of these circuits in two specialists, weakly electric fish and barn owls, has uncovered common design principles and illuminated some aspects of their evolution.

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14. Computing with Self-Excitatory Cliques: A Model and an Application to Hyperacuity-Scale Computation in Visual Cortex

Douglas A. Miller, Steven W. Zucker

https://doi.org/10.1162/089976699300016782 | pp. 21-66

We present a model of visual computation based on tightly inter-connected cliques of pyramidal cells. It leads to a formal theory of cell assemblies, a specific relationship between correlated firing patterns and abstract functionality, and a direct calculation relating estimates of cortical cell counts to orientation hyperacuity. Our network architecture is unique in that (1) it supports a mode of computation that is both reliable and efficient; (2) the current-spike relations are modeled as an analog dynamical system in which the requisite computations can take place on the time scale required for an early stage of visual processing; and (3) the dynamics are triggered by the spatiotemporal response of cortical cells. This final point could explain why moving stimuli improve vernier sensitivity.

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15. Complex Response to Periodic Inhibition in Simple and Detailed Neuronal Models

Corrado Bernasconi, Kaspar Schindler, Ruedi Stoop, Rodney J. Douglas

https://doi.org/10.1162/089976699300016791 | pp. 67-74

Constant current injection with superimposed periodic inhibition gives rise to phase locking as well as chaotic activity in rat neocortical neurons. Here we compare the behavior of a leaky integrate-and-fire neural model with that of a biophysically realistic model of the rat neuron to determine which membrane properties influence the response to such stimuli. We find that only the biophysical model with voltage-sensitive conductances can produce chaotic behavior.

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16. Neuronal Tuning: To Sharpen or Broaden?

Kechen Zhang, Terrence J. Sejnowski

https://doi.org/10.1162/089976699300016809 | pp. 75-84

Sensory and motor variables are typically represented by a population of broadly tuned neurons. A coarser representation with broader tuning can often improve coding accuracy, but sometimes the accuracy may also improve with sharper tuning. The theoretical analysis here shows that the relationship between tuning width and accuracy depends crucially on the dimension of the encoded variable. A general rule is derived for how the Fisher information scales with the tuning width, regardless of the exact shape of the tuning function, the probability distribution of spikes, and allowing some correlated noise between neurons. These results demonstrate a universal dimensionality effect in neural population coding.

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17. Narrow Versus Wide Tuning Curves: What's Best for a Population Code?

Alexandre Pouget, Sophie Denève, Jean-Christophe Ducom, Peter E. Latham

https://doi.org/10.1162/089976699300016818 | pp. 85-90

Neurophysiologists are often faced with the problem of evaluating the quality of a code for a sensory or motor variable, either to relate it to the performance of the animal in a simple discrimination task or to compare the codes at various stages along the neuronal pathway. One common belief that has emerged from such studies is that sharpening of tuning curves improves the quality of the code, although only to a certain point; sharpening beyond that is believed to be harmful. We show that this belief relies on either problematic technical analysis or improper assumptions about the noise. We conclude that one cannot tell, in the general case, whether narrow tuning curves are better than wide ones; the answer depends critically on the covariance of the noise. The same conclusion applies to other manipulations of the tuning curve profiles such as gain increase.

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18. The Effect of Correlated Variability on the Accuracy of a Population Code

L. F. Abbott, Peter Dayan

https://doi.org/10.1162/089976699300016827 | pp. 91-101

We study the impact of correlated neuronal firing rate variability on the accuracy with which an encoded quantity can be extracted from a population of neurons. Contrary to widespread belief, correlations in the variabilities of neuronal firing rates do not, in general, limit the increase in coding accuracy provided by using large populations of encoding neurons. Furthermore, in some cases, but not all, correlations improve the accuracy of a population code.

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19. A Neural Network Model of Temporal Code Generation and Position-Invariant Pattern Recognition

Dean V. Buonomano, Michael M. Merzenich

https://doi.org/10.1162/089976699300016836 | pp. 103-116

Numerous studies have suggested that the brain may encode information in the temporal firing pattern of neurons. However, little is known regarding how information may come to be temporally encoded and about the potential computational advantages of temporal coding. Here, it is shown that local inhibition may underlie the temporal encoding of spatial images. As a result of inhibition, the response of a given cell can be significantly modulated by stimulus features outside its own receptive field. Feedforward and lateral

inhibition can modulate both the firing rate and temporal features, such as latency. In this article, it is shown that a simple neural network model can use local inhibition to generate temporal codes of handwritten numbers. The temporal encoding of a spatial pattern has the interesting and computationally beneficial feature of exhibiting position invariance. This work demonstrates a manner by which the nervous system may generate temporal codes and shows that temporal encoding can be used to create position-invariant codes.

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20. Probabilistic Synaptic Transmission in the Associative Net

Bruce Graham, David Willshaw

https://doi.org/10.1162/089976699300016845 | pp. 117-137

The associative net model of heteroassociative memory with binary-valued synapses has been extended to include recent experimental data indicating that in the hippocampus, one form of synaptic modification is a change in the probability of synaptic transmission. Pattern pairs are stored in the net by a version of the Hebbian learning rule that changes the probability of transmission at synapses where the presynaptic and post-synaptic units are simultaneously active from a low, base value to a high, modified value. Numerical calculations of the expected recall response of this stochastic associative net have been used to assess the performance for different values of the base and modified probabilities. If there is a cost incurred with generating the difference between these probabilities, then a difference of about 0. 4 is optimal. This corresponds to the magnitude of change seen experimentally. Performance can be greatly enhanced by using multiple cue presentations during recall.

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21. A Stochastic Self-Organizing Map for Proximity Data

Thore Graepel, Klaus Obermayer

https://doi.org/10.1162/089976699300016854 | pp. 139-155

We derive an efficient algorithm for topographic mapping of proximity data (TMP), which can be seen as an extension of Kohonen's self-organizing map to arbitrary distance measures. The TMP cost function is derived in a Baysian framework of folded Markov chains for the description of autoencoders. It incorporates the data by a dissimilarity matrix D and the topographic neighborhood by a matrix H of transition probabilities. From the principle of maximum entropy, a nonfactorizing Gibbs distribution is obtained, which is approximated in a mean-field fashion. This allows for maximum likelihood estimation using an expectation-maximization algorithm. In analogy to the transition from topographic vector quantization to the self-organizing map, we suggest an approximation to TMP that is computationally more efficient. In order to prevent convergence to local minima, an annealing scheme in the temperature parameter is introduced, for which the critical temperature of the first phase transition is calculated in terms of D and H. Numerical results demonstrate the working of the algorithm and confirm the analytical results. Finally, the algorithm is used to generate a connection map of areas of the cat's cerebral cortex.

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22. High-Order Contrasts for Independent Component Analysis

J.-F. Cardoso

https://doi.org/10.1162/089976699300016863 | pp. 157-192

This article considers high-order measures of independence for the independent component analysis problem and discusses the class of Jacobi algorithms for their optimization. Several implementations are discussed. We compare the proposed approaches with gradient-based techniques from the algorithmic point of view and also on a set of biomedical data.

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23. Variational Learning in Nonlinear Gaussian Belief Networks

Brendan J. Frey, Geoffrey E. Hinton

https://doi.org/10.1162/089976699300016872 | pp. 193-213

We view perceptual tasks such as vision and speech recognition as inference problems where the goal is to estimate the posterior distribution over latent variables (e.g., depth in stereo vision) given the sensory input. The recent flurry of research in independent component analysis exemplifies the importance of inferring the continuous-valued latent variables of input data. The latent variables found by this method are linearly related to the input, but perception requires nonlinear inferences such as classification and depth estimation. In this article, we present a unifying framework for stochastic neural networks with nonlinear latent variables. Nonlinear units are obtained by passing the outputs of linear gaussian units through various nonlinearities. We present a general variational method that maximizes a lower bound on the likelihood of a training set and give results on two visual feature extraction problems. We also show how the variational method can be used for pattern classification and compare the performance of these nonlinear networks with other methods on the problem of handwritten digit recognition.

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24. Propagating Distributions Up Directed Acyclic Graphs

Eric B. Baum, Warren D. Smith

https://doi.org/10.1162/089976699300016881 | pp. 215-227

In a previous article, we considered game trees as graphical models. Adopting an evaluation function that returned a probability distribution over values likely to be taken at a given position, we described how to build a model of uncertainty and use it for utility-directed growth of the search tree and for deciding on a move after search was completed. In some games, such as chess and Othello, the same position can occur more than once, collapsing the game tree to a directed acyclic graph (DAG). This induces correlations among the distributions at sibling nodes. This article discusses some issues that arise in extending our algorithms to a DAG. We give a simply described algorithm for correctly propagating distributions up a game DAG, taking account of dependencies induced by the DAG structure. This algorithm is exponential time in the worst case. We prove that it is #P complete to propagate distributions up a game DAG correctly. We suggest how our exact propagation algorithm can yield a fast but inexact heuristic.

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25. Modeling and Prediction of Human Behavior

Alex Pentland, Andrew Liu

https://doi.org/10.1162/089976699300016890 | pp. 229-242

We propose that many human behaviors can be accurately described as a set of dynamic modes (e.g., Kalman filters) sequenced together by a Markov chain. We then use these dynamic Markov models to recognize human behaviors from sensory data and to predict human behaviors over a few seconds time. To test the power of this modeling approach, we report an experiment in which we were able to achieve 95% accuracy at predicting automobile drivers' subsequent actions from their initial preparatory movements.

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26. Analog VLSI-Based Modeling of the Primate Oculomotor System

Timothy K. Horiuchi, Christof Koch

https://doi.org/10.1162/089976699300016908 | pp. 243-265

One way to understand a neurobiological system is by building a simulacrum that replicates its behavior in real time using similar constraints. Analog very large-scale integrated (VLSI) electronic circuit technology provides such an enabling technology. We here describe a neuromorphic system that is part of a long-term effort to understand the primate oculomotor system. It requires both fast sensory processing and fast motor control to interact with the world. A one-dimensional hardware model of the primate eye has been built that simulates the physical dynamics of the biological system. It is driven by two different analog VLSI chips, one mimicking cortical visual processing for target selection and tracking and another modeling brain stem circuits that drive the eye muscles. Our oculomotor plant demonstrates both smooth pursuit movements, driven by a retinal velocity error signal, and saccadic eye movements, controlled by retinal position error, and can reproduce several behavioral, stimulation, lesion, and adaptation experiments performed on primates.

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27. JPEG Quality Transcoding Using Neural Networks Trained with a Perceptual Error Measure

John Lazzaro, John Wawrzynek

https://doi.org/10.1162/089976699300016917 | pp. 267-296

A JPEG Quality Transcoder (JQT) converts a JPEG image file that was encoded with low image quality to a larger JPEG image file with reduced visual artifacts, without access to the original uncompressed image. In this article, we describe technology for JQT design that takes a pattern recognition approach to the problem, using a database of images to train statistical models of the artifacts introduced through JPEG compression. In the training procedure for these models, we use a model of human visual perception as an error measure. Our current prototype system removes 32.2% of the artifacts introduced by moderate compression, as measured on an independent test database of linearly coded images using a perceptual error metric. This improvement results in an average PSNR reduction of 0.634 dB.

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28. A Unifying Review of Linear Gaussian Models

Sam T. Roweis, Zoubin Ghahramani

https://doi.org/10.1162/089976699300016674 | pp. 305-345

Factor analysis, principal component analysis, mixtures of gaussian clusters, vector quantization, Kalman filter models, and hidden Markov models can all be unified as variations of unsupervised learning under a

single basic generative model. This is achieved by collecting together disparate observations and derivations made by many previous authors and introducing a new way of linking discrete and continuous state models using a simple nonlinearity. Through the use of other nonlinearities, we show how independent component analysis is also a variation of the same basic generative model. We show that factor analysis and mixtures of gaussians can be implemented in autoencoder neural networks and learned using squared error plus the same regularization term. We introduce a new model for static data, known as sensible principal component analysis, as well as a novel concept of spatially adaptive observation noise. We also review some of the literature involving global and local mixtures of the basic models and provide pseudocode for inference and learning for all the basic models.

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29. Implicit Learning in 3D Object Recognition: The Importance of Temporal Context

Suzanna Becker

https://doi.org/10.1162/089976699300016683 | pp. 347-374

A novel architecture and set of learning rules for cortical self-organization is proposed. The model is based on the idea that multiple information channels can modulate one another's plasticity. Features learned from bottom-up information sources can thus be influenced by those learned from contextual pathways, and vice versa. A maximum likelihood cost function allows this scheme to be implemented in a biologically feasible, hierarchical neural circuit. In simulations of the model, we first demonstrate the utility of temporal context in modulating plasticity. The model learns a representation that categorizes people's faces according to identity, independent of viewpoint, by taking advantage of the temporal continuity in image sequences. In a second set of simulations, we add plasticity to the contextual stream and explore variations in the architecture. In this case, the model learns a two-tiered representation, starting with a coarse view-based clustering and proceeding to a finer clustering of more specific stimulus features. This model provides a tenable account of how people may perform 3D object recognition in a hierarchical, bottom-up fashion.

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30. Relation Between Retinotopical and Orientation Maps in Visual Cortex

Udo Ernst, Klaus Pawelzik, Misha Tsodyks, Terrence J. Sejnowski

https://doi.org/10.1162/089976699300016692 | pp. 375-379

A recent study of cat visual cortex reported abrupt changes in the positions of the receptive fields of adjacent neurons whose preferred orientations strongly differed (Das & Gilbert, 1997). Using a simple cortical model, we show that this covariation of discontinuities in maps of orientation preference and local distortions in maps of visual space reflects collective effects of the lateral cortical feedback.

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31. A Parallel Noise-Robust Algorithm to Recover Depth Information from Radial Flow Fields

Florentin Wörgötter, Alex Cozzi, V. Gerdes

https://doi.org/10.1162/089976699300016700 | pp. 381-416

A parallel algorithm operating on the units ('neurons') of an artificial retina is proposed to recover depth information in a visual scene from radial flow fields induced by ego motion along a given axis. The system consists of up to 600 radii with fewer than 65 radially arranged neurons on each radius. Neurons are connected only to their nearest neighbors, and they are excited as soon as a sufficiently strong gray-level change occurs. The time difference of two subsequently activated neurons is then used by the last-excited neuron to compute the depth information. All algorithmic calculations remain strictly local, and information is exchanged only between adjacent active neurons (except for the final read-out). This, in principle, permits parallel implementation. Furthermore, it is demonstrated that the calculation of the object coordinates requires only a single multiplication with a constant, which is dependent on only the retinal position of the active neuron. The initial restriction to local operations makes the algorithm very noise sensitive. In order to solve this problem, a predication mechanism is introduced. After an object coordinate has been determined, the active neuron computes the time when the next neuronal excitation should take place. This estimated time is transferred to the respective next neuron, which will wait for this excitation only within a certain time window. If the excitation fails to arrive within this window, the previously computed object coordinate is regarded as noisy and discarded. We will show that this predictive mechanism relies also on only a (second) single multiplication with another neuron-dependent constant. Thus, computational complexity remains low, and noisy depth coordinates are efficiently eliminated. Thus, the algorithm is very fast and operates in real time on 128 x 128 images even in a serial implementation on a relatively slow computer. The algorithm is tested on scenes of growing complexity, and a detailed error analysis is provided showing that the depth error remains very low in most cases. A comparison to standard flow-field analysis shows that our algorithm outperforms the older method by far. The analysis of the algorithm also shows that it is generally applicable despite its restrictions, because it is fast and accurate enough such that a complete depth percept can be composed from radial flow field segments. Finally, we suggest how to generalize the algorithm, waiving the restriction of radial flow.

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32. Independent Component Analysis Using an Extended Infomax Algorithm for Mixed Subgaussian and Supergaussian Sources

Te-Won Lee, Mark Girolami, Terrence J. Sejnowski

https://doi.org/10.1162/089976699300016719 | pp. 417-441

An extension of the infomax algorithm of Bell and Sejnowski (1995) is presented that is able blindly to separate mixed signals with sub- and supergaussian source distributions. This was achieved by using a simple type of learning rule first derived by Girolami (1997) by choosing negentropy as a projection pursuit index. Parameterized probability distributions that have sub- and supergaussian regimes were used to derive a general learning rule that preserves the simple architecture proposed by Bell and Sejnowski (1995), is optimized using the natural gradient by Amari (1998), and uses the stability analysis of Cardoso and Laheld (1996) to switch between sub- and supergaussian regimes. We demonstrate that the extended infomax algorithm is able to separate 20 sources with a variety of source distributions easily. Applied to high-dimensional data from electroencephalographic recordings, it is effective at separating artifacts such as eye blinks and line noise from weaker electrical signals that arise from sources in the brain.

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33. Mixtures of Probabilistic Principal Component Analyzers

Michael E. Tipping, Chris Bishop

https://doi.org/10.1162/089976699300016728 | pp. 443-482

Principal component analysis (PCA) is one of the most popular techniques for processing, compressing, and visualizing data, although its effectiveness is limited by its global linearity. While nonlinear variants of PCA have been proposed, an alternative paradigm is to capture data complexity by a combination of local linear PCA projections. However, conventional PCA does not correspond to a probability density, and so there is no unique way to combine PCA models. Therefore, previous attempts to formulate mixture models for PCA have been ad hoc to some extent. In this article, PCA is formulated within a maximum likelihood framework, based on a specific form of gaussian latent variable model. This leads to a well-defined mixture model for probabilistic principal component analyzers, whose parameters can be determined using an expectation-maximization algorithm. We discuss the advantages of this model in the context of clustering, density modeling, and local dimensionality reduction, and we demonstrate its application to image compression and handwritten digit recognition.

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34. Boosted Mixture of Experts: An Ensemble Learning Scheme

Ran Avnimelech, Nathan Intrator

https://doi.org/10.1162/089976699300016737 | pp. 483-497

We present a new supervised learning procedure for ensemble machines, in which outputs of predictors, trained on different distributions, are combined by a dynamic classifier combination model. This procedure may be viewed as either a version of mixture of experts (Jacobs, Jordan, Nowlan, & Hintnon, 1991), applied to classification, or a variant of the boosting algorithm (Schapire, 1990). As a variant of the mixture of experts, it can be made appropriate for general classification and regression problems by initializing the partition of the data set to different experts in a boostlike manner. If viewed as a variant of the boosting algorithm, its main gain is the use of a dynamic combination model for the outputs of the networks. Results are demonstrated on a synthetic example and a digit recognition task from the NIST database and compared with classifical ensemble approaches.

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35. Boosting Regression Estimators

Ran Avnimelech, Nathan Intrator

https://doi.org/10.1162/089976699300016746 | pp. 499-520

There is interest in extending the boosting algorithm (Schapire, 1990) to fit a wide range of regression problems. The threshold-based boosting algorithm for regression used an analogy between classification errors and big errors in regression. We focus on the practical aspects of this algorithm and compare it to other attempts to extend boosting to regression. The practical capabilities of this model are demonstrated on the laser data from the Santa Fe times-series competition and the Mackey-Glass time series, where the results surpass those of standard ensemble average.

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36. An On-Line Agglomerative Clustering Method for Nonstationary Data

Isaac David Guedalia, Mickey London, Michael Werman

https://doi.org/10.1162/089976699300016755 | pp. 521-540

An on-line agglomerative clustering algorithm for nonstationary data is described. Three issues are addressed. The first regards the temporal aspects of the data. The clustering of stationary data by the proposed algorithm is comparable to the other popular algorithms tested (batch and on-line). The second issue addressed is the number of clusters required to represent the data. The algorithm provides an efficient framework to determine the natural number of clusters given the scale of the problem. Finally, the proposed algorithm implicitly minimizes the local distortion, a measure that takes into account clusters with relatively small mass. In contrast, most existing on-line clustering methods assume stationarity of the data. When used to cluster nonstationary data, these methods fail to generate a good representation. Moreover, most current algorithms are computationally intensive when determining the correct number of clusters. These algorithms tend to neglect clusters of small mass due to their minimization of the global distortion (Energy).

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37. Hidden Neural Networks

Anders Krogh, Søren Riis

https://doi.org/10.1162/089976699300016764 | pp. 541-563

A general framework for hybrids of hidden Markov models (HMMs) and neural networks (NNs) called hidden neural networks (HNNs) is described. The article begins by reviewing standard HMMs and estimation by conditional maximum likelihood, which is used by the HNN. In the HNN, the usual HMM probability parameters are replaced by the outputs of state-specific neural networks. As opposed to many other hybrids, the HNN is normalized globally and therefore has a valid probabilistic interpretation. All parameters in the HNN are estimated simultaneously according to the discriminative conditional maximum likelihood criterion. The HNN can be viewed as an undirected probabilistic independence network (a graphical model), where the neural networks provide a compact representation of the clique functions. An evaluation of the HNN on the task of recognizing broad phoneme classes in the TIMIT database shows clear performance gains compared to standard HMMs tested on the same task.

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38. A Cerebellar Model of Timing and Prediction in the Control of Reaching

Andrew G. Barto, Andrew H. Fagg, N. Sitkoff, James C. Houk

https://doi.org/10.1162/089976699300016575 | pp. 565-594

A simplified model of the cerebellum was developed to explore its potential for adaptive, predictive control based on delayed feedback information. An abstract representation of a single Purkinje cell with multistable properties was interfaced, using a formalized premotor network, with a simulated single degree-of-freedom limb. The limb actuator was a nonlinear spring-mass system based on the nonlinear velocity dependence of the stretch reflex. By including realistic mossy fiber signals, as well as realistic conduction delays in afferent and efferent pathways, the model allowed the investigation of timing and predictive processes relevant to cerebellar involvement in the control of movement. The model regulates movement by learning to react in an anticipatory fashion to sensory feedback. Learning depends on training information generated from corrective movements and uses a temporally asymmetric form of plasticity for the parallel fiber synapses on Purkinje cells.

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39. Improved Multidimensional Scaling Analysis Using Neural Networks with Distance-Error Backpropagation

Luís Garrido, Sergio Gómez, Jaume Roca

https://doi.org/10.1162/089976699300016584 | pp. 595-600

We show that neural networks, with a suitable error function for backpropagation, can be successfully used for metric multidimensional scaling (MDS) (i.e., dimensional reduction while trying to preserve the original distances between patterns) and are in fact able to outdo the standard algebraic approach to MDS, known as classical scaling.

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40. Firing Rate Distributions and Efficiency of Information Transmission of Inferior Temporal Cortex Neurons to Natural Visual Stimuli

Alessandro Treves, Stefano Panzeri, Edmund T. Rolls, Michael C. A. Booth, Edward A. Wakeman

https://doi.org/10.1162/089976699300016593 | pp. 601-631

The distribution of responses of sensory neurons to ecological stimulation has been proposed to be designed to maximize information transmission, which according to a simple model would imply an exponential distribution of spike counts in a given time window. We have used recordings from inferior temporal cortex neurons responding to quasi-natural visual stimulation (presented using a video of everyday lab scenes and a large number of static images of faces and natural scenes) to assess the validity of this exponential model and to develop an alternative simple model of spike count distributions. We find that the exponential model has to be rejected in 84% of cases (at the $p < 0.01$ level). A new model, which accounts for the firing rate distribution found in terms of slow and fast variability in the inputs that produce neuronal activation, is rejected statistically in only 16% of cases. Finally, we show that the neurons are moderately efficient at transmitting information but not optimally efficient.

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41. Collective Behavior of Networks with Linear (VLSI) Integrate-and-Fire Neurons

Stefano Fusi, Maurizio Mattia

https://doi.org/10.1162/089976699300016601 | pp. 633-652

We analyze in detail the statistical properties of the spike emission process of a canonical integrate-and-fire neuron, with a linear integrator and a lower bound for the depolarization, as often used in VLSI implementations (Mead, 1989). The spike statistics of such neurons appear to be qualitatively similar to conventional (exponential) integrate-and-fire neurons, which exhibit a wide variety of characteristics observed in cortical recordings. We also show that, contrary to current opinion, the dynamics of a network composed of such neurons has two stable fixed points, even in the purely excitatory network, corresponding to two different states of reverberating activity. The analytical results are compared with numerical simulations and are found to be in good agreement.

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42. Recurrent Sampling Models for the Helmholtz Machine

Peter Dayan

https://doi.org/10.1162/089976699300016610 | pp. 653-677

Many recent analysis-by-synthesis density estimation models of cortical learning and processing have made the crucial simplifying assumption that units within a single layer are mutually independent given the states of units in the layer below or the layer above. In this article, we suggest using either a Markov random field or an alternative stochastic sampling architecture to capture explicitly particular forms of dependence within each layer. We develop the architectures in the context of real and binary Helmholtz machines. Recurrent sampling can be used to capture correlations within layers in the generative or the recognition models, and we also show how these can be combined.

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43. Feature Extraction Through LOCOCODE

Sepp Hochreiter, Jürgen Schmidhuber

https://doi.org/10.1162/089976699300016629 | pp. 679-714

Low-complexity coding and decoding (LOCOCODE) is a novel approach to sensory coding and unsupervised learning. Unlike previous methods, it explicitly takes into account the information-theoretic complexity of the code generator. It computes lococodes that convey information about the input data and can be computed and decoded by low-complexity mappings. We implement LOCOCODE by training autoassociators with flat minimum search, a recent, general method for discovering low-complexity neural nets. It turns out that this approach can unmix an unknown number of independent data sources by extracting a minimal number of low-complexity features necessary for representing the data. Experiments show that unlike codes obtained with standard autoencoders, lococodes are based on feature detectors, never unstructured, usually sparse, and sometimes factorial or local (depending on statistical properties of the data). Although LOCOCODE is not explicitly designed to enforce sparse or factorial codes, it extracts optimal codes for difficult versions of the "bars" benchmark problem, whereas independent component analysis (ICA) and principal component analysis (PCA) do not. It produces familiar, biologically plausible feature detectors when applied to real-world images and codes with fewer bits per pixel than ICA and PCA. Unlike ICA, it does not need to know the number of independent sources. As a preprocessor for a vowel recognition benchmark problem, it sets the stage for excellent classification performance. Our results reveal an interesting, previously ignored connection between two important fields: regularizer research and ICA-related research. They may represent a first step toward unification of regularization and unsupervised learning.

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44. Discontinuities in Recurrent Neural Networks

Ricard Gavaldà, Hava T. Siegelmann

https://doi.org/10.1162/089976699300016638 | pp. 715-745

This article studies the computational power of various discontinuous real computational models that are based on the classical analog recurrent neural network (ARNN). This ARNN consists of finite number of neurons; each neuron computes a polynomial net function and a sigmoid-like continuous activation function. We introduce arithmetic networks as ARNN augmented with a few simple discontinuous (e.g., threshold or zero test) neurons. We argue that even with weights restricted to polynomial time computable reals, arithmetic networks are able to compute arbitrarily complex recursive functions. We identify many types of

neural networks that are at least as powerful as arithmetic nets, some of which are not in fact discontinuous, but they boost other arithmetic operations in the net function (e.g., neurons that can use divisions and polynomial net functions inside sigmoid-like continuous activation functions). These arithmetic networks are equivalent to the Blum-Shub-Smale model, when the latter is restricted to a bounded number of registers. With respect to implementation on digital computers, we show that arithmetic networks with rational weights can be simulated with exponential precision, but even with polynomial-time computable real weights, arithmetic networks are not subject to any fixed precision bounds. This is in contrast with the ARNN that are known to demand precision that is linear in the computation time. When nontrivial periodic functions (e.g., fractional part, sine, tangent) are added to arithmetic networks, the resulting networks are computationally equivalent to a massively parallel machine. Thus, these highly discontinuous networks can solve the presumably intractable class of PSPACE-complete problems in polynomial time.

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45. Parameter Convergence and Learning Curves for Neural Networks

Terrence L. Fine, Sayandev Mukherjee

https://doi.org/10.1162/089976699300016647 | pp. 747-769

We revisit the oft-studied asymptotic (in sample size) behavior of the parameter or weight estimate returned by any member of a large family of neural network training algorithms. By properly accounting for the characteristic property of neural networks that their empirical and generalization errors possess multiple minima, we rigorously establish conditions under which the parameter estimate converges strongly into the set of minima of the generalization error. Convergence of the parameter estimate to a particular value cannot be guaranteed under our assumptions. We then evaluate the asymptotic distribution of the distance between the parameter estimate and its nearest neighbor among the set of minima of the generalization error. Results on this question have appeared numerous times and generally assert asymptotic normality, the conclusion expected from familiar statistical arguments concerned with maximum likelihood estimators. These conclusions are usually reached on the basis of somewhat informal calculations, although we shall see that the situation is somewhat delicate. The preceding results then provide a derivation of learning curves for generalization and empirical errors that leads to bounds on rates of convergence.

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46. Analog Neural Nets with Gaussian or Other Common Noise Distributions Cannot Recognize Arbitrary Regular Languages

Wolfgang Maass, Eduardo D. Sontag

https://doi.org/10.1162/089976699300016656 | pp. 771-782

We consider recurrent analog neural nets where the output of each gate is subject to gaussian noise or any other common noise distribution that is nonzero on a sufficiently large part of the state-space. We show that many regular languages cannot be recognized by networks of this type, and we give a precise characterization of languages that can be recognized. This result implies severe constraints on possibilities for constructing recurrent analog neural nets that are robust against realistic types of analog noise. On the other hand, we present a method for constructing feedfor-ward analog neural nets that are robust with regard to analog noise of this type.

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47. Discriminant Component Pruning: Regularization and Interpretation of Multilayered Backpropagation Networks

Randal A. Koene, Yoshio Takane

https://doi.org/10.1162/089976699300016665 | pp. 783-802

Neural networks are often employed as tools in classification tasks. The use of large networks increases the likelihood of the task's being learned, although it may also lead to increased complexity. Pruning is an effective way of reducing the complexity of large networks. We present discriminant components pruning (DCP), a method of pruning matrices of summed contributions between layers of a neural network. Attempting to interpret the underlying functions learned by the network can be aided by pruning the network. Generalization performance should be maintained at its optimal level following pruning. We demonstrate DCP's effectiveness at maintaining generalization performance, applicability to a wider range of problems, and the usefulness of such pruning for network interpretation. Possible enhancements are discussed for the identification of the optimal reduced rank and inclusion of nonlinear neural activation functions in the pruning algorithm.

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48. Independent Factor Analysis

Hagai Attias

https://doi.org/10.1162/089976699300016458 | pp. 803-851

We introduce the independent factor analysis (IFA) method for recovering independent hidden sources from their observed mixtures. IFA generalizes and unifies ordinary factor analysis (FA), principal component analysis (PCA), and independent component analysis (ICA), and can handle not only square noiseless mixing but also the general case where the number of mixtures differs from the number of sources and the data are noisy. IFA is a two-step procedure. In the first step, the source densities, mixing matrix, and noise covariance are estimated from the observed data by maximum likelihood. For this purpose we present an expectation-maximization (EM) algorithm, which performs unsupervised learning of an associated probabilistic model of the mixing situation. Each source in our model is described by a mixture of gaussians; thus, all the probabilistic calculations can be performed analytically. In the second step, the sources are reconstructed from the observed data by an optimal nonlinear estimator. A variational approximation of this algorithm is derived for cases with a large number of sources, where the exact algorithm becomes intractable. Our IFA algorithm reduces to the one for ordinary FA when the sources become gaussian, and to an EM algorithm for PCA in the zero-noise limit. We derive an additional EM algorithm specifically for noiseless IFA. This algorithm is shown to be superior to ICA since it can learn arbitrary source densities from the data. Beyond blind separation, IFA can be used for modeling multidimensional data by a highly constrained mixture of gaussians and as a tool for nonlinear signal encoding.

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49. A Fast, Compact Approximation of the Exponential Function

Nicol N. Schraudolph

https://doi.org/10.1162/089976699300016467 | pp. 853-862

Neural network simulations often spend a large proportion of their time computing exponential functions. Since the exponentiation routines of typical math libraries are rather slow, their replacement with a fast

approximation can greatly reduce the overall computation time. This article describes how exponentiation can be approximated by manipulating the components of a standard (IEEE-754) floating-point representation. This models the exponential function as well as a lookup table with linear interpolation, but is significantly faster and more compact.

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50. On Cross Validation for Model Selection

Isabelle Rivals, L. Personnaz

https://doi.org/10.1162/089976699300016476 | pp. 863-870

In response to Zhu and Rower (1996), a recent communication (Goutte, 1997) established that leave-one-out cross validation is not subject to the "no-free-lunch" criticism. Despite this optimistic conclusion, we show here that cross validation has very poor performances for the selection of linear models as compared to classic statistical tests. We conclude that the statistical tests are preferable to cross validation for linear as well as for nonlinear model selection.

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51. Analysis of Integrate-and-Fire Neurons: Synchronization of Synaptic Input and Spike Output

Anthony N. Burkitt, G. M. Clark

https://doi.org/10.1162/089976699300016485 | pp. 871-901

A new technique for analyzing the probability distribution of output spikes for the integrate-and-fire model is presented. This technique enables us to investigate models with arbitrary synaptic response functions that incorporate both leakage across the membrane and a rise time of the postsynaptic potential. The results, which are compared with numerical simulations, are exact in the limit of a large number of small-amplitude inputs. This method is applied to the synchronization problem, in which we examine the relationship between the spread in arrival times of the inputs (the temporal jitter of the synaptic input) and the resultant spread in the times at which the output spikes are generated (output jitter). The results of previous studies, which indicated that the ration of the output jitter to the input jitter is consistently less than one and that it decreases for increasing numbers of inputs, are confirmed for three classes of the integrate-and-fire model. In addition to the previously identified factors of axonal propagation times and synaptic jitter, we identify the variation in the spike-generating thresholds of the neurons and the variation in the number of active inputs as being important factors that determine the timing jitter in layered networks. Previously observed phase differences between optimally and suboptimally stimulated neurons may be understood in terms of the relative time taken to reach threshold.

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52. Dynamic Stochastic Synapses as Computational Units

Wolfgang Maass, Anthony M. Zador

https://doi.org/10.1162/089976699300016494 | pp. 903-917

In most neural network models, synapses are treated as static weights that change only with the slow time scales of learning. It is well known, however, that synapses are highly dynamic and show use-dependent plasticity over a wide range of time scales. Moreover, synaptic transmission is an inherently stochastic

process: a spike arriving at a presynaptic terminal triggers the release of a vesicle of neurotransmitter from a release site with a probability that can be much less than one. We consider a simple model for dynamic stochastic synapses that can easily be integrated into common models for networks of integrate-and-fire neurons (spiking neurons). The parameters of this model have direct interpretations in terms of synaptic physiology. We investigate the consequences of the model for computing with individual spikes and demonstrate through rigorous theoretical results that the computational power of the network is increased through the use of dynamic synapses.

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53. Spatiotemporal Coding in the Cortex: Information Flow-Based Learning in Spiking Neural Networks

Gustavo Deco, Bernd Schürmann

https://doi.org/10.1162/089976699300016502 | pp. 919-934

We introduce a learning paradigm for networks of integrate-and-fire spiking neurons that is based on an information-theoretic criterion. This criterion can be viewed as a first principle that demonstrates the experimentally observed fact that cortical neurons display synchronous firing for some stimuli and not for others. The principle can be regarded as the postulation of a nonparametric reconstruction method as optimization criteria for learning the required functional connectivity that justifies and explains synchronous firing for binding of features as a mechanism for spatiotemporal coding. This can be expressed in an information-theoretic way by maximizing the discrimination ability between different sensory inputs in minimal time.

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54. The Ornstein-Uhlenbeck Process Does Not Reproduce Spiking Statistics of Neurons in Prefrontal Cortex

Shigeru Shinomoto, Yutaka Sakai, Shintaro Funahashi

https://doi.org/10.1162/089976699300016511 | pp. 935-951

Cortical neurons of behaving animals generate irregular spike sequences. Recently, there has been a heated discussion about the origin of this irregularity. Softky and Koch (1993) pointed out the inability of standard single-neuron models to reproduce the irregularity of the observed spike sequences when the model parameters are chosen within a certain range that they consider to be plausible. Shadlen and Newsome (1994), on the other hand, demonstrated that a standard leaky integrate-and-fire model can reproduce the irregularity if the inhibition is balanced with the excitation. Motivated by this discussion, we attempted to determine whether the Ornstein-Uhlenbeck process, which is naturally derived from the leaky integration assumption, can in fact reproduce higher-order statistics of biological data. For this purpose, we consider actual neuronal spike sequences recorded from the monkey prefrontal cortex to calculate the higher-order statistics of the interspike intervals. Consistency of the data with the model is examined on the basis of the coefficient of variation and the skewness coefficient, which are, respectively, a measure of the spiking irregularity and a measure of the asymmetry of the interval distribution. It is found that the biological data are not consistent with the model if the model time constant assumes a value within a certain range believed to cover all reasonable values. This fact suggests that the leaky integrate-and-fire model with the assumption of uncorrelated inputs is not adequate to account for the spiking in at least some cortical neurons.

55. Random Neural Networks with Multiple Classes of Signals

Erol Gelenbe, Jean■Michel Fourneau

https://doi.org/10.1162/089976699300016520 | pp. 953-963

By extending the pulsed recurrent random neural network (RNN) discussed in Gelenbe (1989, 1990, 1991), we propose a recurrent random neural network model in which each neuron processes several distinctly characterized streams of "signals" or data. The idea that neurons may be able to distinguish between the pulses they receive and use them in a distinct manner is biologically plausible. In engineering applications, the need to process different streams of information simultaneously is commonplace (e.g., in image processing, sensor fusion, or parallel processing systems). In the model we propose, each distinct stream is a class of signals in the form of spikes. Signals may arrive to a neuron from either the outside world (exogenous signals) or other neurons (endogenous signals). As a function of the signals it has received, a neuron can fire and then send signals of some class to another neuron or to the outside world. We show that the multiple signal class random model with exponential interfiring times, Poisson external signal arrivals, and Markovian signal movements between neurons has product form; this implies that the distribution of its state (i.e., the probability that each neuron of the network is excited) can be computed simply from the solution of a system of 2Cn simultaneous nonlinear equations where C is the number of signal classes and n is the number of neurons. Here we derive the stationary solution for the multiple class model and establish necessary and sufficient conditions for the existence of the stationary solution. The recurrent random neural network model with multiple classes has already been successfully applied to image texture generation (Atalay & Gelenbe, 1992), where multiple signal classes are used to model different colors in the image.

56. An Adaptive Bayesian Pruning for Neural Networks in a Non-Stationary Environment

John Sum, Chi-Sing Leung, Gilbert H. Young, Laiwan Chan, Wing■Kay Kan

https://doi.org/10.1162/089976699300016539 | pp. 965-976

Pruning a neural network to a reasonable smaller size, and if possible to give a better generalization, has long been investigated. Conventionally the common technique of pruning is based on considering error sensitivity measure, and the nature of the problem being solved is usually stationary. In this article, we present an adaptive pruning algorithm for use in a nonstationary environment. The idea relies on the use of the extended Kalman filter (EKF) training method. Since EKF is a recursive Bayesian algorithm, we define a weight-importance measure in term of the sensitivity of a posteriori probability. Making use of this new measure and the adaptive nature of EKF, we devise an adaptive pruning algorithm called adaptive Bayesian pruning. Simulation results indicate that in a noisy nonstationary environment, the proposed pruning algorithm is able to remove network redundancy adaptively and yet preserve the same generalization ability.

57. Pruning Using Parameter and Neuronal Metrics

Piërre van de Laar, Tom Heskes

https://doi.org/10.1162/089976699300016548 | pp. 977-993

In this article, we introduce a measure of optimality for architecture selection algorithms for neural networks: the distance from the original network to the new network in a metric defined by the probability distributions of all possible networks. We derive two pruning algorithms, one based on a metric in parameter space and the other based on a metric in neuron space, which are closely related to well-known architecture selection algorithms, such as GOBS. Our framework extends the theoretically range of validity of GOBS and therefore can explain results observed in previous experiments. In addition, we give some computational improvements for these algorithms.

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58. No Free Lunch for Early Stopping

Zehra Çataltepe, Yaser S. Abu■Mostafa, Malik Magdon■Ismail

https://doi.org/10.1162/089976699300016557 | pp. 995-1009

We show that with a uniform prior on models having the same training error, early stopping at some fixed training error above the training error minimum results in an increase in the expected generalization error.

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59. Blind Separation of a Mixture of Uniformly Distributed Source Signals: A Novel Approach

Jayanta Kumar Basak, ■■■-■■■ ■■■■■

https://doi.org/10.1162/089976699300016566 | pp. 1011-1034

A new, efficient algorithm for blind separation of uniformly distributed sources is proposed. The mixing matrix is assumed to be orthogonal by prewhitening the observed signals. The learning rule adaptively estimates the mixing matrix by conceptually rotating a unit hypercube so that all output signal components are contained within or on the hypercube. Under some ideal constraints, it has been theoretically shown that the algorithm is very similar to an ideal O(1/T2) convergent algorithm, which is much faster than the existing O(1/T) convergent algorithms. The algorithm has been generalized to take care of the noisy signals by adaptively dilating the hypercube in conjunction with its rotation.

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60. Comparison of Approximate Methods for Handling Hyperparameters

David Mackay

https://doi.org/10.1162/089976699300016331 | pp. 1035-1068

I examine two approximate methods for computational implementation of Bayesian hierarchical models, that is, models that include unknown hyperparameters such as regularization constants and noise levels. In the evidence framework, the model parameters are integrated over, and the resulting evidence is maximized over the hyperparameters. The optimized hyperparameters are used to define a gaussian approximation to the posterior distribution. In the alternative MAP method, the true posterior probability is found by integrating over the hyperparameters. The true posterior is then maximized over the model parameters, and a gaussian approximation is made. The similarities of the two approaches and their relative merits are discussed, and comparisons are made with the ideal hierarchical Bayesian solution. In moderately ill-posed problems, integration over hyperparameters yields a probability distribution with a skew peak, which causes signifi-cant biases to arise in the MAP method. In contrast, the evidence framework is shown to introduce negligible

predictive error under straightforward conditions. General lessons are drawn concerning inference in many dimensions.

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61. Relating the Slope of the Activation Function and the Learning Rate Within a Recurrent Neural Network

Danilo P. Mandic, Jonathon A. Chambers

https://doi.org/10.1162/089976699300016340 | pp. 1069-1077

A relationship between the learning rate η in the learning algorithm, and the slope β in the nonlinear activation function, for a class of recurrent neural networks (RNNs) trained by the real-time recurrent learning algorithm is provided. It is shown that an arbitrary RNN can be obtained via the referent RNN, with some deterministic rules imposed on its weights and the learning rate. Such relationships reduce the number of degrees of freedom when solving the nonlinear optimization task of finding the optimal RNN parameters.

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62. Network Stability from Activity-Dependent Regulation of Neuronal Conductances

Jorge Golowasch, Michael A. Casey, L. F. Abbott, Eve Marder

https://doi.org/10.1162/089976699300016359 | pp. 1079-1096

Activity-dependent plasticity appears to play an important role in the modification of neurons and neural circuits that occurs during development and learning. Plasticity is also essential for the maintenance of stable patterns of activity in the face of variable environmental and internal conditions. Previous theoretical and experimental results suggest that neurons stabilize their activity by altering the number or characteristics of ion channels to regulate their intrinsic electrical properties. We present both experimental and modeling evidence to show that activity-dependent regulation of conductances, operating at the level of individual neurons, can also stabilize network activity. These results indicate that the stomatogastric ganglion of the crab can generate a characteristic rhythmic pattern of activity in two fundamentally different modes of operation. In one mode, the rhythm is strictly conditional on the presence of neuromodulatory afferents from adjacent ganglia. In the other, it is independent of neuromodulatory input but relies on newly developed intrinsic properties of the component neurons.

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63. Optimal Detection of Flash Intensity Differences Using Rod Photocurrent Observations

Peter N. Steinmetz, Raimond L. Winslow

https://doi.org/10.1162/089976699300016368 | pp. 1097-1111

The rod photocurrent contains two noise components that may limit the detectability of flash intensity increments. The limits imposed by the low- and high-frequency noise components were assessed by computing the performance of an optimal detector of increments in flash intensity. The limits imposed by these noise components depend on the interval of observation of the photocurrent signal. When the entire photocurrent signal, lasting 3 or more seconds, is observed, the low-frequency component of the

photocurrent noise (attributed to the quantal noise of the incoming light, as well as random isomerizations of enzymes within the phototransduction cascade) is the most significant limitation on detectability. When only the first 380 ms or less is observed, the high-frequency component of the noise (due to the thermal isomerizations of the cGMP-gated channel) presents a significant limit on the detectability of flashes.

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64. On the Role of Biophysical Properties of Cortical Neurons in Binding and Segmentation of Visual Scenes

Paul F. M. J. Verschure, Peter König

https://doi.org/10.1162/089976699300016377 | pp. 1113-1138

Neuroscience is progressing vigorously, and knowledge at different levels of description is rapidly accumulating. To establish relationships between results found at these different levels is one of the central challenges. In this simulation study, we demonstrate how microscopic cellular properties, taking the example of the action of modulatory substances onto the membrane leakage current, can provide the basis for the perceptual functions reflected in the macroscopic behavior of a cortical network. In the first part, the action of the modulatory system on cortical dynamics is investigated. First, it is demonstrated that the inclusion of these biophysical properties in a model of the primary visual cortex leads to the dynamic formation of synchronously active neuronal assemblies reflecting a context-dependent binding and segmentation of image components. Second, it is shown that the differential regulation of the leakage current can be used to bias the interactions of multiple cortical modules. This allows the flexible use of different feature domains for scene segmentation. Third, we demonstrate how, within the proposed architecture, the mapping of a moving stimulus onto the spatial dimension of the network results in an increased speed of synchronization. In the second part, we demonstrate how the differential regulation of neuromodulatory activity can be achieved in a self-consistent system. Three different mechanisms are described and investigated. This study thus demonstrates how a modulatory system, affecting the biophysical properties of single cells, can be used to achieve context-dependent processing at the system level.

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65. The Continuum of Operating Modes for a Passive Model Neuron

Michael A. Kisley, George L. Gerstein

https://doi.org/10.1162/089976699300016386 | pp. 1139-1154

Whether cortical neurons act as coincidence detectors or temporal integrators has implications for the way in which the cortex encodes information--by average firing rate or by precise timing of action potentials. In this study, we examine temporal coding by a simple passive-membrane model neuron responding to a full spectrum of multisynaptic input patterns, from highly coincident to temporally dispersed. The temporal precision of the model's action potentials varies continuously along the spectrum, depends very little on the number of synaptic inputs, and is shown to be tightly correlated with the mean slope of the membrane potential preceding the output spikes. These results are shown to be largely independent of the size of postsynaptic potentials, of random background synaptic activity, and of shape of the correlated multisynaptic input pattern. An experimental test involving membrane potential slope is suggested to help determine the basic operating mode of an observed cortical neuron.

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66. Structure Learning in Conditional Probability Models via an Entropic Prior and Parameter Extinction

Matthew Brand

https://doi.org/10.1162/089976699300016395 | pp. 1155-1182

We introduce an entropic prior for multinomial parameter estimation problems and solve for its maximum a posteriori (MAP) estimator. The prior is a bias for maximally structured and minimally ambiguous models. In conditional probability models with hidden state, iterative MAP estimation drives weakly supported parameters toward extinction, effectively turning them off. Thus, structure discovery is folded into parameter estimation. We then establish criteria for simplifying a probabilistic model's graphical structure by trimming parameters and states, with a guarantee that any such deletion will increase the posterior probability of the model. Trimming accelerates learning by sparsifying the model. All operations monotonically and maximally increase the posterior probability, yielding structure-learning algorithms only slightly slower than parameter estimation via expectation-maximization and orders of magnitude faster than search-based structure induction. When applied to hidden Markov model training, the resulting models show superior generalization to held-out test data. In many cases the resulting models are so sparse and concise that they are interpretable, with hidden states that strongly correlate with meaningful categories.

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67. On the Approximation Rate of Hierarchical Mixtures-of-Experts for Generalized Linear Models

Wenxin Jiang, Martin A. Tanner

https://doi.org/10.1162/089976699300016403 | pp. 1183-1198

We investigate a class of hierarchical mixtures-of-experts (HME) models where generalized linear models with nonlinear mean functions of the form psi (alpha + xT beta) are mixed. Here psi (.) is the inverse link function. It is shown that mixtures of such mean functions can approximate a class of smooth functions of the form psi (h(x)), where h(.) epsilon W2;K infinity (a Sobolev class over [0, 1]s), as the number of experts m in the network increases. An upper bound of the approximation rate is given as O(m-2/s) in Lp norm. This rate can be achieved within the family of HME structures with no more than s-layers, where s is the dimension of the predictor x.

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68. Stochastic Learning of Strategic Equilibria for Auctions

Samy Bengio, Yoshua Bengio, Jacques Robert, Gilles Bélanger

https://doi.org/10.1162/089976699300016412 | pp. 1199-1209

This article presents a new application of stochastic adaptive learning algorithms to the computation of strategic equilibria in auctions. The proposed approach addresses the problems of tracking a moving target and balancing exploration (of action space) versus exploitation (of better modeled regions of action space). Neural networks are used to represent a stochastic decision model for each bidder. Experiments confirm the correctness and usefulness of the approach.

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69. Kalman Filter Implementation of Self-Organizing Feature Maps

Karin Haese

https://doi.org/10.1162/089976699300016421 | pp. 1211-1233

The self-organizing learning algorithm of Kohonen and most of its extensions are controlled by two learning parameters, the learning coefficient and the width of the neighborhood function, which have to be chosen empirically because neither rules nor methods for their calculation exist. Consequently, often time-consuming parameter studies precede neighborhood-preserving feature maps of the learning data. To circumvent those lengthy numerical studies, this article describes the learning process by a state-space model in order to use the linear Kalman filter algorithm training the feature maps. Then the Kalman filter equations calculate the learning coefficient online during the training, while the width of the neighborhood function needs to be estimated by a second extended Kalman filter for the process of neighborhood preservation. The performance of the Kalman filter implementation is demonstrated on toy problems as well as on a crab classification problem. The results of crab classification are compared to those of generative topographic mapping, an alternative method to the self-organizing feature map.

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70. A Fast Histogram-Based Postprocessor That Improves Posterior Probability Estimates

Wei Wei, Todd K. Leen, Etienne Barnard

https://doi.org/10.1162/089976699300016430 | pp. 1235-1248

Although the outputs of neural network classifiers are often considered to be estimates of posterior class probabilities, the literature that assesses the calibration accuracy of these estimates illustrates that practical networks often fall far short of being ideal estimators. The theorems used to justify treating network outputs as good posterior estimates are based on several assumptions: that the network is sufficiently complex to model the posterior distribution accurately, that there are sufficient training data to specify the network, and that the optimization routine is capable of finding the global minimum of the cost function. Any or all of these assumptions may be violated in practice. This article does three things. First, we apply a simple, previously used histogram technique to assess graphically the accuracy of posterior estimates with respect to individual classes. Second, we introduce a simple and fast remapping procedure that transforms network outputs to provide better estimates of posteriors. Third, we use the remapping in a real-world telephone speech recognition system. The remapping results in a 10% reduction of both word-level error rates (from 4.53% to 4.06%) and sentence-level error rates (from 16.38% to 14.69%) on one corpus, and a 29% reduction at sentence-level error (from 6.3% to 4.5%) on another. The remapping required negligible additional overhead (in terms of both parameters and calculations). McNemar's test shows that these levels of improvement are statistically significant.

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71. Training a Sigmoidal Node Is Hard

Don Hush

https://doi.org/10.1162/089976699300016449 | pp. 1249-1260

This article proves that the task of computing near-optimal weights for sigmoidal nodes under the L 1 regression norm is NP-Hard. For the special case where the sigmoid is piecewise linear, we prove a slightly stronger result: that computing the optimal weights is NP-Hard. These results parallel that for the one-node

pattern recognition problem—that determining the optimal weights for a threshold logic node is also intractable. Our results have important consequences for constructive algorithms that build a regression model one node at a time. It suggests that although such methods are (in principle) capable of producing efficient size representations (Barron, 1993; Jones, 1992), finding such representations may be computationally intractable. These results holds only in the deterministic sense; that is, they do not exclude the possibility that such representations may be found efficiently with high probability. In fact it motivates the use of heuristic or randomized algorithms for this problem.

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72. Seeing White: Qualia in the Context of Decoding Population Codes

Sidney R. Lehky, Terrence J. Sejnowski

https://doi.org/10.1162/089976699300016232 | pp. 1261-1280

When the nervous system is presented with multiple simultaneous inputs of some variable, such as wavelength or disparity, they can be combined to give rise to qualitatively new percepts that cannot be produced by any single input value. For example, there is no single wavelength that appears white. Many models of decoding neural population codes have problems handling multiple inputs, either attempting to extract a single value of the input parameter or, in some cases, registering the presence of multiple inputs without synthesizing them into something new. These examples raise a more general issue regarding the interpretation of population codes. We propose that population decoding involves not the extraction of specific values of the physical inputs, but rather a transformation from the input space to some abstract representational space that is not simply related to physical parameters. As a specific example, a four-layer network is presented that implements a transformation from wavelength to a high-level hue-saturation color space.

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73. Adaptive Calibration of Imaging Array Detectors

Marco Budinich, Renato Frison

https://doi.org/10.1162/089976699300016241 | pp. 1281-1296

We present two methods for nonuniformity correlation of imaging array detectors based on neural networks; both exploit image properties to supply lack of calibrations and maximize the entropy of the output. The first method uses a self-organizing net that produces a linear correction of the raw data with coefficients that adapt continuously. The second method employs a kind of contrast equalization curve to match pixel distributions. Our work originates from silicon detectors, but the treatment is general enough to be applicable to many kinds of array detectors like those used in infrared imaging or in high-energy physics.

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74. Modeling the Combination of Motion, Stereo, and Vergence Angle Cues to Visual Depth

Ione Fine, Robert A. Jacobs

https://doi.org/10.1162/089976699300016250 | pp. 1297-1330

Three models of visual cue combination were simulated: a weak fusion model, a modified weak model, and a strong model. Their relative strengths and weaknesses are evaluated on the basis of their performances

on the tasks of judging the depth and shape of an ellipse. The models differ in the amount of interaction that they permit among the cues of stereo, motion, and vergence angle. Results suggest that the constrained nonlinear interaction of the modified weak model allows better performance than either the linear interaction of the weak model or the unconstrained nonlinear interaction of the strong model. Further examination of the modified weak model revealed that its weighting of motion and stereo cues was dependent on the task, the viewing distance, and, to a lesser degree, the noise model. Although the dependencies were sensible from a computational viewpoint, they were sometimes inconsistent with psychophysical experimental data. In a second set of experiments, the modified weak model was given contradictory motion and stereo information. One cue was informative in the sense that it indicated an ellipse, while the other cue indicated a flat surface. The modified weak model rapidly reweighted its use of stereo and motion cues as a function of each cue's informativeness. Overall, the simulation results suggest that relative to the weak and strong models, the modified weak fusion model is a good candidate model of the combination of motion, stereo, and vergence angle cues, although the results also highlight areas in which this model needs modification or further elaboration.

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75. Inferring Perceptual Saliency Fields from Viewpoint-Dependent Recognition Data

Florin Cutzu, Michael J. Tarr

https://doi.org/10.1162/089976699300016269 | pp. 1331-1348

We present an algorithm for computing the relative perceptual saliencies of the features of a three-dimensional object using either goodness-of-view scores measured at several viewpoints or perceptual similarities among several object views. This technique addresses the inverse, ill-posed version of the direct problem of predicting goodness-of-view scores or viewpoint similarities when the object features are known. On the basis of a linear model for the direct problem, we solve the inverse problem using the method of regularization. The critical assumption we make to regularize the solution is that perceptual salience varies slowly on the surface of the object. The salient regions derived using this assumption empirically indicate what object structures are important in human three-dimensional object perception, a domain where theories typically have been based on somewhat ad hoc features.

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76. Backward Projections in the Cerebral Cortex: Implications for Memory Storage

Alfonso Renart, Néstor Parga, Edmund T. Rolls

https://doi.org/10.1162/089976699300016278 | pp. 1349-1388

Cortical areas are characterized by forward and backward connections between adjacent cortical areas in a processing stream. Within each area there are recurrent collateral connections between the pyramidal cells. We analyze the properties of this architecture for memory storage and processing. Hebb-like synaptic modifiability in the connections and attractor states are incorporated. We show the following: (1) The number of memories that can be stored in the connected modules is of the same order of magnitude as the number that can be stored in any one module using the recurrent collateral connections, and is proportional to the number of effective connections per neuron. (2) Cooperation between modules leads to a small increase in memory capacity. (3) Cooperation can also help retrieval in a module that is cued with a noisy or incomplete

pattern. (4) If the connection strength between modules is strong, then global memory states that reflect the pairs of patterns on which the modules were trained together are found. (5) If the intermodule connection strengths are weaker, then separate, local memory states can exist in each module. (6) The boundaries between the global and local retrieval states, and the nonretrieval state, are delimited. All of these properties are analyzed quantitatively with the techniques of statistical physics.

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77. The Relationship Between Synchronization Among Neuronal Populations and Their Mean Activity Levels

D. Chawla, Erik D. Lumer, Karl Friston

https://doi.org/10.1162/089976699300016287 | pp. 1389-1411

In the past decade the importance of synchronized dynamics in the brain has emerged from both empirical and theoretical perspectives. Fast dynamic synchronous interactions of an oscillatory or nonoscillatory nature may constitute a form of temporal coding that underlies feature binding and perceptual synthesis. The relationship between synchronization among neuronal populations and the population firing rates addresses two important issues: the distinction between rate coding and synchronization coding models of neuronal interactions and the degree to which empirical measurements of population activity, such as those employed by neuroimaging, are sensitive to changes in synchronization. We examined the relationship between mean population activity and synchronization using biologically plausible simulations. In this article, we focus on continuous stationary dynamics. (In a companion article, Chawla (forthcoming), we address the same issue using stimulus-evoked transients.) By manipulation parameters such as extrinsic input, intrinsic noise, synaptic efficacy, density of extrinsic connections, the voltage-sensitive nature of postsynaptic mechanisms, the number of neurons, and the laminar structure within the populations, we were able to introduce variations in both mean activity and synchronization under a variety of simulated neuronal architectures. Analyses of the simulated spike trains and local field potentials showed that in nearly every domain of the model's parameter space, mean activity and synchronization were tightly coupled. This coupling appears to be mediated by an increase in synchronous gain when effective membrane time constants are lowered by increased activity. These observations show that under the assumptions implicit in our models, rate coding and synchrony coding in neural systems with reciprocal interconnections are two perspectives on the same underlying dynamic. This suggests that in the absence of specific mechanisms decoupling changes in synchronization from firing levels, indexes of brain activity that are based purely on synaptic activity (e.g., functional magnetic resonance imaging) may also be sensitive to changes in synchronous coupling.

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78. Fast Calculation of Short-Term Depressing Synaptic Conductances

Michèle Giugliano, Marco Bove, M. Grattarola

https://doi.org/10.1162/089976699300016296 | pp. 1413-1426

An efficient implementation of synaptic transmission models in realistic network simulations is an important theme of computational neuroscience. The amount of CPU time required to simulate synaptic interactions can increase as the square of the number of units of such networks, depending on the connectivity convergence. As a consequence, any realistic description of synaptic phenomena, incorporating biophysical details, is computationally highly demanding. We present a consolidating algorithm based on a biophysical extended model of ligand-gated postsynaptic channels, describing short-term plasticity such as synaptic

depression. The considerable speed-up of simulation times makes this algorithm suitable for investigating emergent collective effects of short-term depression in large-scale networks of model neurons.

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79. Algorithmic Stability and Sanity-Check Bounds for Leave-One-Out Cross-Validation

Michael Kearns, Dana Ron

https://doi.org/10.1162/089976699300016304 | pp. 1427-1453

In this article we prove sanity-check bounds for the error of the leave-one-out cross-validation estimate of the generalization error: that is, bounds showing that the worst-case error of this estimate is not much worse than that of the training error estimate. The name sanity check refers to the fact that although we often expect the leave-one-out estimate to perform considerably better than the training error estimate, we are here only seeking assurance that its performance will not be considerably worse. Perhaps surprisingly, such assurance has been given only for limited cases in the prior literature on cross-validation. Any nontrivial bound on the error of leave-one-out must rely on some notion of algorithmic stability. Previous bounds relied on the rather strong notion of hypothesis stability, whose application was primarily limited to nearest-neighbor and other local algorithms. Here we introduce the new and weaker notion of error stability and apply it to obtain sanity-check bounds for leave-one-out for other classes of learning algorithms, including training error minimization procedures and Bayesian algorithms. We also provide lower bounds demonstrating the necessity of some form of error stability for proving bounds on the error of the leave-one-out estimate, and the fact that for training error minimization algorithms, in the worst case such bounds must still depend on the Vapnik-Chervonenkis dimension of the hypothesis class.

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80. Convergence Properties of the Softassign Quadratic Assignment Algorithm

Anand Rangarajan, Alan Yuille, Eric Mjolsness

https://doi.org/10.1162/089976699300016313 | pp. 1455-1474

The softassign quadratic assignment algorithm is a discrete-time, continuous-state, synchronous updating optimizing neural network. While its effectiveness has been shown in the traveling salesman problem, graph matching, and graph partitioning in thousands of simulations, its convergence properties have not been studied. Here, we construct discrete-time Lyapunov functions for the cases of exact and approximate doubly stochastic constraint satisfaction, which show convergence to a fixed point. The combination of good convergence properties and experimental success makes the softassign algorithm an excellent choice for neural quadratic assignment optimization.

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81. Learning to Design Synergetic Computers with an Extended Symmetric Diffusion Network

Koji Okuhara, Shunji Osaki, Masaaki Kijima

https://doi.org/10.1162/089976699300016322 | pp. 1475-1491

This article proposes an extended symmetric diffusion network that is applied to the design of synergetic computers. The state of a synergetic computer is translated to that of order parameters whose dynamics is described by a stochastic differential equation. The order parameter converges to the Boltzmann distribution, under some condition on the drift term, derived by the Fokker-Planck equation. The network can learn the dynamics of the order parameters from a nonlinear potential. This property is necessary to design the coefficient values of the synergetic computer. We propose a searching function for the image processing executed by the synergetic computer. It is shown that the image processing with the searching function is superior to the usual image-associative function of synergetic computation. The proposed network can be related, as a special case, to the discrete-state Boltzmann machine by some transformation. Finally, the extended symmetric diffusion network is applied to the estimation problem of an entire density function, as well as the proposed searching function for the image processing.

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82. Prediction Games and Arcing Algorithms

Leo Breiman

https://doi.org/10.1162/089976699300016106 | pp. 1493-1517

The theory behind the success of adaptive reweighting and combining algorithms (arcing) such as Adaboost (Freund & Schapire, 1996a, 1997) and others in reducing generalization error has not been well understood. By formulating prediction as a game where one player makes a selection from instances in the training set and the other a convex linear combination of predictors from a finite set, existing arcing algorithms are shown to be algorithms for finding good game strategies. The minimax theorem is an essential ingredient of the convergence proofs. An arcing algorithm is described that converges to the optimal strategy. A bound on the generalization error for the combined predictors in terms of their maximum error is proven that is sharper than bounds to date. Schapire, Freund, Bartlett, and Lee (1997) offered an explanation of why Adaboost works in terms of its ability to produce generally high margins. The empirical comparison of Adaboost to the optimal arching algorithm shows that their explanation is not complete.

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83. Can Hebbian Volume Learning Explain Discontinuities in Cortical Maps?

Graeme Mitchison, Nicholas V. Swindale

https://doi.org/10.1162/089976699300016115 | pp. 1519-1526

It has recently been shown that orientation and retinotopic position, both of which are mapped in primary visual cortex, can show correlated jumps (Das & Gilbert, 1997). This is not consistent with maps generated by Kohonen's algorithm (Kohonen, 1982), where changes in mapped variables tend to be anticorrelated. We show that it is possible to obtain correlated jumps by introducing a Hebbian component (Hebb, 1949) into Kohonen's algorithm. This correspondents to a volume learning mechanism where synaptic facilitation depends not only on the spread of a signal from a maximally active neuron but also requires postsynaptic activity at a synapse. The maps generated by this algorithm show discontinuities across which both orientation and retinotopic position change rapidly, but these regions, which include the orientation singularities, are also aligned with the edges of ocular dominance columns, and this is not a realistic feature of cortical maps. We conclude that cortical maps are better modeled by standard, non-Hebbian volume learning, perhaps coupled with some other mechanism (e.g., that of Ernst, Pawelzik, Tsodyks, & Sejnowski, 1999) to produce receptive field shifts.

84. Disambiguating Different Covariation Types

Carlos D. Brody

https://doi.org/10.1162/089976699300016124 | pp. 1527-1535

Covariations in neuronal latency or excitability can lead to peaks in spike train covariograms that may be very similar to those caused by spike timing synchronization (see companion article). Two quantitative methods are described here. The first is a method to estimate the excitability component of a covariogram, based on trial-by-trial estimates of excitability. Once estimated, this component may be subtracted from the covariogram, leaving only other types of contributions. The other is a method to determine whether the covariogram could potentially have been caused by latency covariations.

85. Correlations Without Synchrony

Carlos D. Brody

https://doi.org/10.1162/089976699300016133 | pp. 1537-1551

Peaks in spike train correlograms are usually taken as indicative of spike timing synchronization between neurons. Strictly speaking, however, a peak merely indicates that the two spike trains were not independent. Two biologically plausible ways of departing from independence that are capable of generating peaks very similar to spike timing peaks are described here: covariations over trials in response latency and covariations over trials in neuronal excitability. Since peaks due to these interactions can be similar to spike timing peaks, interpreting a correlogram may be a problem with ambiguous solutions. What peak shapes do latency or excitability interactions generate? When are they similar to spike timing peaks? When can they be ruled out from having caused an observed correlogram peak? These are the questions addressed here. The previous article in this issue proposes quantitative methods to tell cases apart when latency or excitability covariations cannot be ruled out.

86. On Decoding the Responses of a Population of Neurons from Short Time Windows

Stefano Panzeri, Alessandro Treves, Simon R. Schultz, Edmund T. Rolls

https://doi.org/10.1162/089976699300016142 | pp. 1553-1577

The effectiveness of various stimulus identification (decoding) procedures for extracting the information carried by the responses of a population of neurons to a set of repeatedly presented stimuli is studied analytically, in the limit of short time windows. It is shown that in this limit, the entire information content of the responses can sometimes be decoded, and when this is not the case, the lost information is quantified. In particular, the mutual information extracted by taking into account only the most likely stimulus in each trial turns out to be, if not equal, much closer to the true value than that calculated from all the probabilities that each of the possible stimuli in the set was the actual one. The relation between the mutual information extracted by decoding and the percentage of correct stimulus decodings is also derived analytically in the same limit, showing that the metric content index can be estimated reliably from a few cells recorded from brief periods. Computer simulations as well as the activity of real neurons recorded in the primate hippocampus serve to confirm these results and illustrate the utility and limitations of the approach.

87. Short-Term Synaptic Plasticity and Network Behavior

Werner M. Kistler, J. Leo van Hemmen

https://doi.org/10.1162/089976699300016151 | pp. 1579-1594

We develop a minimal time-continuous model for use-dependent synaptic short-term plasticity that can account for both short-term depression and short-term facilitation. It is analyzed in the context of the spike response neuron model. Explicit expressions are derived for the synaptic strength as a function of previous spike arrival times. These results are then used to investigate the behavior of large networks of highly interconnected neurons in the presence of short-term synaptic plasticity. We extend previous results so as to elucidate the existence and stability of limit cycles with coherently firing neurons. After the onset of an external stimulus, we have found complex transient network behavior that manifests itself as a sequence of different modes of coherent firing until a stable limit cycle is reached.

88. Synchrony and Desynchrony in Integrate-and-Fire Oscillators

Shannon R. Campbell, DeLiang L. Wang, C. Jayaprakash

https://doi.org/10.1162/089976699300016160 | pp. 1595-1619

Due to many experimental reports of synchronous neural activity in the brain, there is much interest in understanding synchronization in networks of neural oscillators and its potential for computing perceptual organization. Contrary to Hopfield and Herz (1995), we find that networks of locally coupled integrate-and-fire oscillators can quickly synchronize. Furthermore, we examine the time needed to synchronize such networks. We observe that these networks synchronize at times proportional to the logarithm of their size, and we give the parameters used to control the rate of synchronization. Inspired by locally excitatory globally inhibitory oscillator network (LEGION) dynamics with relaxation oscillators (Terman & Wang, 1995), we find that global inhibition can play a similar role of desynchronization in a network of integrate-and-fire oscillators. We illustrate that a LEGION architecture with integrate-and-fire oscillators can be similarly used to address image analysis.

89. Fast Global Oscillations in Networks of Integrate-and-Fire Neurons with Low Firing Rates

Nicolas Brunel, Vincent Hakim

https://doi.org/10.1162/089976699300016179 | pp. 1621-1671

We study analytically the dynamics of a network of sparsely connected inhibitory integrate-and-fire neurons in a regime where individual neurons emit spikes irregularly and at a low rate. In the limit when the number of neurons --> infinity, the network exhibits a sharp transition between a stationary and an oscillatory global activity regime where neurons are weakly synchronized. The activity becomes oscillatory when the inhibitory feedback is strong enough. The period of the global oscillation is found to be mainly controlled by synaptic times but depends also on the characteristics of the external input. In large but finite networks, the analysis shows that global oscillations of finite coherence time generically exist both above and below the critical inhibition threshold. Their characteristics are determined as functions of systems parameters in these two different regions. The results are found to be in good agreement with numerical simulations.

90. Concentration Tuning Mediated by Spare Receptor Capacity in Olfactory Sensory Neurons: A Theoretical Study

Thomas A. Cleland, Christiane Linster

https://doi.org/10.1162/089976699300016188 | pp. 1673-1690

The olfactory system is capable of detecting odorants at very low concentrations. Physiological experiments have demonstrated odorant sensitivities down to the picomolar range in preparations from the sensory epithelium. However, the contemporary model for olfactory signal transduction provides that odorants bind to olfactory receptors with relatively low specificity and consequently low affinity, making this detection of low-concentration odorants theoretically difficult to understand. We employ a computational model to demonstrate how olfactory sensory neuron (OSN) sensitivity can be tuned by modulation of receptor-effector coupling and/or by other mechanisms regulating spare receptor capacity, thus resolving this conundrum. The EC10-90 intensity tuning ranges (ITRs) of whole olfactory glomeruli and postsynaptic mitral cells are considerably broader than the commensurate ITRs of individual OSNs. These data are difficult to reconcile with certain contemporary hypotheses that convergent OSNs in mammals exhibit a homogeneous population of olfactory receptors and identical tuning for odor stimuli. We show that heterogeneity in spare receptor capacities within a convergent OSN population can increase the ITR (EC10-90) of a convergent population of OSNs regardless of the presence or absence of a diversity of receptor expression within the population. The modulation of receptor-effector coupling has been observed in OSNs; other mechanisms for cellular regulation of spare receptor capacity are also highly plausible (e.g., quantitative regulation of the relative expression levels of receptor and effector proteins). We present a model illustrating that these processes can underlie both how OSNs come to exhibit high sensitivity to odorant stimuli without necessitating increased ligand-receptor binding affinities or specificities and how a population of convergent OSNs could exhibit a broader concentration sensitivity than its individual constituent neurons, even given a population expressing identical odorant receptors. The regulation of spare receptor capacity may play an important role in the olfactory system's ability to reliably detect low odor concentrations, discriminate odor intensities, and segregate this intensity information from representations of odor quality.

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91. A Computational Model for Visual Selection

Yali Amit, Donald Geman

https://doi.org/10.1162/089976699300016197 | pp. 1691-1715

We propose a computational model for detecting and localizing instances from an object class in static gray-level images. We divide detection into visual selection and final classification, concentrating on the former: drastically reducing the number of candidate regions that require further, usually more intensive, processing, but with a minimum of computation and missed detections. Bottom-up processing is based on local groupings of edge fragments constrained by loose geometrical relationships. They have no a priori semantic or geometric interpretation. The role of training is to select special groupings that are moderately likely at certain places on the object but rate in the background. We show that the statistics in both populations are stable. The candidate regions are those that contain global arrangements of several local groupings. Whereas our model was not conceived to explain brain functions, it does cohere with evidence about the functions of neurons in V1 and V2, such as responses to coarse or incomplete patterns (e.g., illusory contours) and to scale and translation invariance in IT. Finally, the algorithm is applied to face and

symbol detection.

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92. Associative Memory in a Multimodular Network

Nir Levy, D. Horn, Eytan Ruppin

https://doi.org/10.1162/089976699300016205 | pp. 1717-1737

Recent imaging studies suggest that object knowledge is stored in the brain as a distributed network of many cortical areas. Motivated by these observations, we study a multimodular associative memory network, whose functional goal is to store patterns with different coding levels--patterns that vary in the number of modules in which they are encoded. We show that in order to accomplish this task, synaptic inputs should be segregated into intramodular projections and intermodular projections, with the latter undergoing additional nonlinear dendritic processing. This segregation makes sense anatomically if the intermodular projections represent distal synaptic connections on apical dendrites. It is then straightforward to show that memories encoded in more modules are more resilient to focal afferent damage. Further hierarchical segregation of intermodular connections on the dendritic tree improves this resilience, allowing memory retrieval from input to just one of the modules in which it is encoded.

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93. Sparse Code Shrinkage: Denoising of Nongaussian Data by Maximum Likelihood Estimation

Aapo Hyvärinen

https://doi.org/10.1162/089976699300016214 | pp. 1739-1768

Sparse coding is a method for finding a representation of data in which each of the components of the representation is only rarely significantly active. Such a representation is closely related to redundancy reduction and independent component analysis, and has some neurophysiological plausibility. In this article, we show how sparse coding can be used for denoising. Using maximum likelihood estimation of nongaussian variables corrupted by gaussian noise, we show how to apply a soft-thresholding (shrinkage) operator on the components of sparse coding so as to reduce noise. Our method is closely related to the method of wavelet shrinkage, but it has the important benefit over wavelet methods that the representation is determined solely by the statistical properties of the data. The wavelet representation, on the other hand, relies heavily on certain mathematical properties (like self-similarity) that may be only weakly related to the properties of natural data.

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94. Improving the Convergence of the Backpropagation Algorithm Using Learning Rate Adaptation Methods

George D. Magoulas, Michael N. Vrahatis, George Androulakis

https://doi.org/10.1162/089976699300016223 | pp. 1769-1796

This article focuses on gradient-based backpropagation algorithms that use either a common adaptive learning rate for all weights or an individual adaptive learning rate for each weight and apply the Goldstein/Armijo line search. The learning-rate adaptation is based on descent techniques and estimates of the local Lipschitz constant that are obtained without additional error function and gradient evaluations. The

proposed algorithms improve the backpropagation training in terms of both convergence rate and convergence characteristics, such as stable learning and robustness to oscillations. Simulations are conducted to compare and evaluate the convergence behavior of these gradient-based training algorithms with several popular training methods.

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95. Detecting and Estimating Signals in Noisy Cable Structures, I: Neuronal Noise Sources

Amit Manwani, Christof Koch

https://doi.org/10.1162/089976699300015972 | pp. 1797-1829

In recent theoretical approaches addressing the problem of neural coding, tools from statistical estimation and information theory have been applied to quantify the ability of neurons to transmit information through their spike outputs. These techniques, though fairly general, ignore the specific nature of neuronal processing in terms of its known biophysical properties. However, a systematic study of processing at various stages in a biophysically faithful model of a single neuron can identify the role of each stage in information transfer. Toward this end, we carry out a theoretical analysis of the information loss of a synaptic signal propagating along a linear, one-dimensional, weakly active cable due to neuronal noise sources along the way, using both a signal reconstruction and a signal detection paradigm. Here we begin such an analysis by quantitatively characterizing three sources of membrane noise: (1) thermal noise due to the passive membrane resistance, (2) noise due to stochastic openings and closings of voltage-gated membrane channels (NA+ and K+), and (3) noise due to random, background synaptic activity. Using analytical expressions for the power spectral densities of these noise sources, we compare their magnitudes in the case of a patch of membrane from a cortical pyramidal cell and explore their dependence on different biophysical parameters.

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96. Detecting and Estimating Signals in Noisy Cable Structures, II: Information Theoretical Analysis

Amit Manwani, Christof Koch

https://doi.org/10.1162/089976699300015981 | pp. 1831-1873

This is the second in a series of articles that seek to recast classical single-neuron biophysics in information-theoretical terms. Classical cable theory focuses on analyzing the voltage or current attenuation of a synaptic signal as it propagates from its dendritic input location to the spike initiation zone. On the other hand, we are interested in analyzing the amount of information lost about the signal in this process due to the presence of various noise sources distributed throughout the neuronal membrane. We use a stochastic version of the linear one-dimensional cable equation to derive closed-form expressions for the second-order moments of the fluctuations of the membrane potential associated with different membrane current noise sources: thermal noise, noise due to the random opening and closing of sodium and potassium channels, and noise due to the presence of "spontaneous" synaptic input. We consider two different scenarios. In the signal estimation paradigm, the time course of the membrane potential at a location on the cable is used to reconstruct the detailed time course of a random, band-limited current injected some distance away. Estimation performance is characterized in terms of the coding fraction and the mutual information. In the signal detection paradigm, the membrane potential is used to determine whether a distant synaptic event

occurred within a given observation interval. In the light of our analytical results, we speculate that the length of weakly active apical dendrites might be limited by the information loss due to the accumulated noise between distal synaptic input sites and the soma and that the presence of dendritic nonlinearities probably serves to increase dendritic information transfer.

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97. Natural Gradient Learning for Over- and Under-Complete Bases in ICA

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https://doi.org/10.1162/089976699300015990 | pp. 1875-1883

Independent component analysis or blind source separation is a new technique of extracting independent signals from mixtures. It is applicable even when the number of independent sources is unknown and is larger or smaller than the number of observed mixture signals. This article extends the natural gradient learning algorithm to be applicable to these overcomplete and undercomplete cases. Here, the observed signals are assumed to be whitened by preprocessing, so that we use the natural Riemannian gradient in Stiefel manifolds.

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98. Combined 5 × 2 cv F Test for Comparing Supervised Classification Learning Algorithms

Ethem Alpaydm
https://doi.org/10.1162/089976699300016007 | pp. 1885-1892

Dietterich (1998) reviews five statistical tests and proposes the 5 x 2 cv t test for determining whether there is a significant difference between the error rates of two classifiers. In our experiments, we noticed that the 5 x 2 cv t test result may vary depending on factors that should not affect the test, and we propose a variant, the combined 5 x 2 cv F test, that combines multiple statistics to get a more robust test. Simulation results show that this combined version of the test has lower type I error and higher power than 5 x 2 cv proper.

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99. Adaptive Neural Coding Dependent on the Time-Varying Statistics of the Somatic Input Current

Jonghan Shin, Christof Koch, Rodney J. Douglas
https://doi.org/10.1162/089976699300016016 | pp. 1893-1913

It is generally assumed that nerve cells optimize their performance to reflect the statistics of their input. Electronic circuit analogs of neurons require similar methods of self-optimization for stable and autonomous operation. We here describe and demonstrate a biologically plausible adaptive algorithm that enables a neuron to adapt the current threshold and the slope (or gain) of its current-frequency relationship to match the man (or dc offset) and variance (or dynamic range or contrast) of the time-varying somatic input current. The adaptation algorithm estimates the somatic current signal from the spike train by way of the intracellular somatic calcium concentration, thereby continuously adjusting the neurons' firing dynamics. This principle is shown to work in an analog VLSI-designed silicon neuron.

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100. A Reinforcement Learning Approach to Online Clustering

Aristidis Likas

https://doi.org/10.1162/089976699300016025 | pp. 1915-1932

A general technique is proposed for embedding online clustering algorithms based on competitive learning in a reinforcement learning framework. The basic idea is that the clustering system can be viewed as a reinforcement learning system that learns through reinforcements to follow the clustering strategy we wish to implement. In this sense, the reinforcement guided competitive learning (RGCL) algorithm is proposed that constitutes a reinforcement-based adaptation of learning vector quantization (LVQ) with enhanced clustering capabilities. In addition, we suggest extensions of RGCL and LVQ that are characterized by the property of sustained exploration and significantly improve the performance of those algorithms, as indicated by experimental tests on well-known data sets.

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101. Replicator Equations, Maximal Cliques, and Graph Isomorphism

Marcello Pelillo

https://doi.org/10.1162/089976699300016034 | pp. 1933-1955

We present a new energy-minimization framework for the graph isomorphism problem that is based on an equivalent maximum clique formulation. The approach is centered around a fundamental result proved by Motzkin and Straus in the mid-1960s, and recently expanded in various ways, which allows us to formulate the maximum clique problem in terms of a standard quadratic program. The attractive feature of this formulation is that a clear one-to-one correspondence exists between the solutions of the quadratic program and those in the original, combinatorial problem. To solve the program we use the so-called replicator equations--a class of straightforward continuous- and discrete-time dynamical systems developed in various branches of theoretical biology. We show how, despite their inherent inability to escape from local solutions, they nevertheless provide experimental results that are competitive with those obtained using more elaborate mean-field annealing heuristics.

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102. Independent Component Analysis: A Flexible Nonlinearity and Decorrelating Manifold Approach

Richard Everson, Stephen Roberts

https://doi.org/10.1162/089976699300016043 | pp. 1957-1983

Independent component analysis (ICA) finds a linear transformation to variables that are maximally statistically independent. We examine ICA and algorithms for finding the best transformation from the point of view of maximizing the likelihood of the data. In particular, we discuss the way in which scaling of the unmixing matrix permits a "static" nonlinearity to adapt to various marginal densities. We demonstrate a new algorithm that uses generalized exponential functions to model the marginal densities and is able to separate densities with light tails. We characterize the manifold of decorrelating matrices and show that it lies along the ridges of high-likelihood unmixing matrices in the space of all unmixing matrices. We show how to find the optimum ICA matrix on the manifold of decorrelating matrices, and as an example we use the algorithm to find independent component basis vectors for an ensemble of portraits.

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103. On the Design of BSB Neural Associative Memories Using Semidefinite Programming

Jooyoung Park, Hyuk Cho, Daihee Park

https://doi.org/10.1162/089976699300016052 | pp. 1985-1994

This article is concerned with the reliable search for optimally performing BSB (brain state in a box) neural associative memories given a set of prototype patterns to be stored as stable equilibrium points. By converting and/or modifying the nonlinear constraints of a known formulation for the synthesis of BSB-based associative memories into linear matrix inequalities, we recast the synthesis into semidefinite programming problems and solve them by recently developed interior point methods. The validity of this approach is illustrated by a design example.

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104. How to Design a Connectionist Holistic Parser

E. K. S. Ho, Lai Wan Chan

https://doi.org/10.1162/089976699300016061 | pp. 1995-2016

Connectionist holistic parsing offers a viable and attractive alternative to traditional algorithmic parsers. With exposure to a limited subset of grammatical sentences and their corresponding parse trees only, a holistic parser is capable of learning inductively the grammatical regularity underlying the training examples that affects the parsing process. In the past, various connectionist parsers have been proposed. Each approach had its own unique characteristics, and yet some techniques were shared in common. In this article, various dimensions underlying the design of a holistic parser are explored, including the methods to encode sentences and parse trees, whether a sentence and its corresponding parse tree share the same representation, the use of confluent inference, and the inclusion of phrases in the training set. Different combinations of these design factors give rise to different holistic parsers. In succeeding discussions, we scrutinize these design techniques and compare the performances of a few parsers on language parsing, including the confluent preorder parser, the backpropagation parsing network, the XERIC parser of Berg (1992), the modular connectionist parser of Sharkey and Sharkey (1992), Reilly's (1992) model, and their derivatives. Experiments are performed to evaluate their generalization capability and robustness. The results reveal a number of issues essential for building an effective holistic parser.

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105. A Unified Analysis of Value-Function-Based Reinforcement-Learning Algorithms

Csaba Szepesvári, Michael L. Littman

https://doi.org/10.1162/089976699300016070 | pp. 2017-2060

Reinforcement learning is the problem of generating optimal behavior in a sequential decision-making environment given the opportunity of interacting with it. Many algorithms for solving reinforcement-learning problems work by computing improved estimates of the optimal value function. We extend prior analyses of reinforcement-learning algorithms and present a powerful new theorem that can provide a unified analysis of such value-function-based reinforcement-learning algorithms. The usefulness of the theorem lies in how it allows the convergence of a complex asynchronous reinforcement-learning algorithm to be proved by verifying that a simpler synchronous algorithm converges. We illustrate the application of the theorem by

analyzing the convergence of Q-learning, model-based reinforcement learning, Q-learning with multistate updates, Q-learning for Markov games, and risk-sensitive reinforcement learning.

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106. Neuronal Regulation: A Mechanism for Synaptic Pruning During Brain Maturation

Gal Chechik, Isaac Meilijson, Eytan Ruppin

https://doi.org/10.1162/089976699300016089 | pp. 2061-2080

Human and animal studies show that mammalian brains undergo massive synaptic pruning during childhood, losing about half of the synapses by puberty. We have previously shown that maintaining the network performance while synapses are deleted requires that synapses be properly modified and pruned, with the weaker synapses removed. We now show that neuronal regulation, a mechanism recently observed to maintain the average neuronal input field of a postsynaptic neuron, results in a weight-dependent synaptic modification. Under the correct range of the degradation dimension and synaptic upper bound, neuronal regulation removes the weaker synapses and judiciously modifies the remaining synapses. By deriving optimal synaptic modification functions in an excitatory-inhibitory network, we prove that neuronal regulation implements near-optimal synaptic modification and maintains the performance of a network undergoing massive synaptic pruning. These findings support the possibility that neural regulation complements the action of Hebbian synaptic changes in the self-organization of the developing brain.

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107. Comparison of SOM Point Densities Based on Different Criteria

Teuvo Kohonen

https://doi.org/10.1162/089976699300016098 | pp. 2081-2095

Point densities of model (codebook) vectors in self-organizing maps (SOMs) are evaluated in this article. For a few one-dimensional SOMs with finite grid lengths and a given probability density function of the input, the numerically exact point densities have been computed. The point density derived from the SOM algorithm turned out to be different from that minimizing the SOM distortion measure, showing that the model vectors produced by the basic SOM algorithm in general do not exactly coincide with the optimum of the distortion measure. A new computing technique based on the calculus of variations has been introduced. It was applied to the computation of point densities derived from the distortion measure for both the classical vector quantization and the SOM with general but equal dimensionality of the input vectors and the grid, respectively. The power laws in the continuum limit obtained in these cases were found to be identical.